A Sample Article Using IEEEtran.cls for IEEE Journals and Transactions

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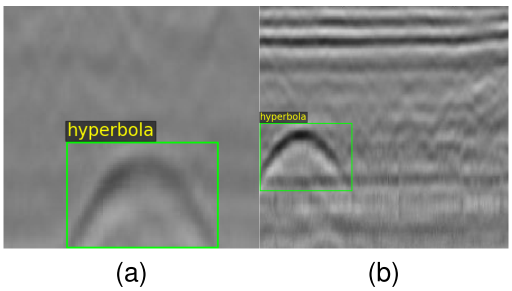

Fig. 1. Hyperbola samples.

***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LaTeX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LaTeX, paper, template, typesetting.**

I. Introduction

THIS file is intended to serve as a "sample article file"

II. Related work

A. Hyperbola recognition task

B. Labelling tools

III. Proposed method

A. Hyperbola annotation tool

We designed a visual annotation tool for hyperbolic ribbon-shaped targets, whose user interface is illustrated in the Fig. 2. First, the user selects an image folder, and the tool automatically loads all images within it. The original image is displayed on the lower-left panel, while the annotated image is shown on the right. A mouse click on the original image marks the vertex position of the hyperbola, and five adjustable parameters are provided on the right panel.

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The five parameters are defined as follows: the horizontal and vertical coordinates of the hyperbola vertex, hyperbola width, height, and ribbon thickness. By clicking the *Add as New Annotation* button, the current parameter configuration is saved as a new annotation for the active image, and multiple annotations can be added to a single image.

All annotations are stored in JSON format with the following structure:

```
{
  "024.jpg": [
    {
      "label": "hyperbola",
      "x_vertex": 127.04,
      "y_vertex": 163.84,
      "width": 87.65,
      "height": 59.76,
      "thickness": 33.03
    },
    {
      "label": "hyperbola",
      "x_vertex": 186.56,
      "y_vertex": 109.76,
      "width": 97.35,
      "height": 79.35,
      "thickness": 33.03
    }
  ]
}
```

Here, *024.jpg* denotes the image filename, and the remaining entries represent the parameters and their corresponding values.

B. Hyperbola detection nerual network

IV. Method

A. Overview

Given a single-channel GPR B-scan $I \in \mathbb{R}^{1\times H\times W}$ (here $H = W = 640$), HyperbolaNet predicts a set of axis-aligned bounding boxes enclosing the hyperbolic reflections. The network comprises three parts: a convolutional encoder backbone, a *supervised attention branch*, and a YOLO-style detection head. The central contribution is the attention branch, whose attention

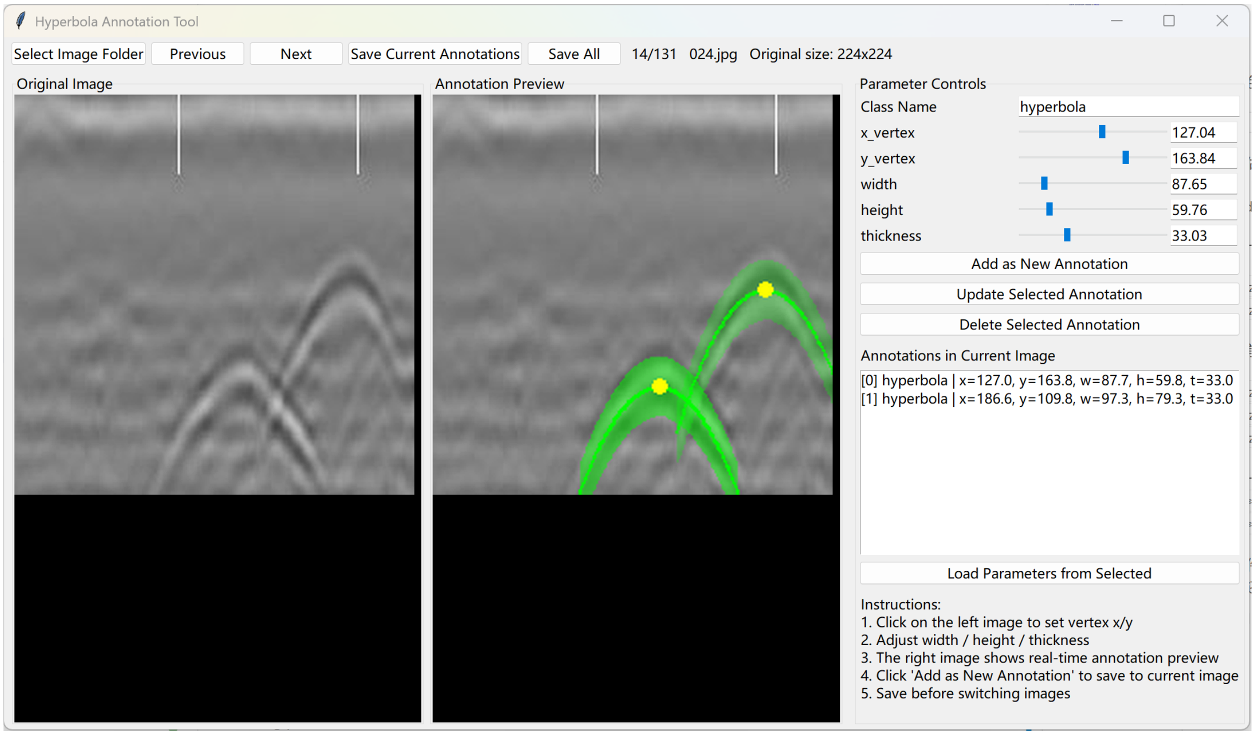


Fig. 2. tool

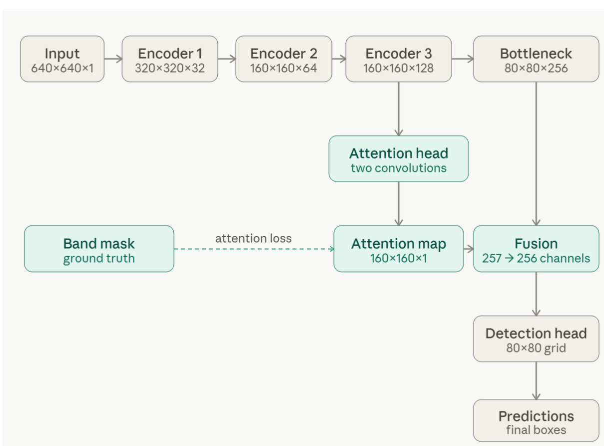


Fig. 3. 11

map is explicitly supervised by a rasterized hyperbola band mask *inside the backbone*, and is then injected into the bottleneck feature by channel concatenation. An overview is shown in Fig. **??**.

B. Encoder backbone

The encoder consists of three down-sampling stages followed by a bottleneck. Each stage is a convolutional block—two 3×3 convolutions, each followed by batch normalization (BN) and ReLU—followed by 2×2 max pooling, so that the channel depth doubles and the spatial resolution halves at each stage. The resulting feature dimensions (channels $\times$ H $\times$ W) are $1 \times 640 \times 640$, $32 \times 320 \times 320$, $64 \times 160 \times 160$, $128 \times 160 \times 160$, and $256 \times 80 \times 80$ at the bottleneck. We denote by $f_3 \in \mathbb{R}^{C_3 \times \frac{H}{4} \times \frac{W}{4}}$ the output of the third stage *before* its pooling ($C_3 = 128$, stride 4), and by $F_b \in \mathbb{R}^{C_b \times \frac{H}{8} \times \frac{W}{8}}$ the bottleneck feature ($C_b = 256$, stride 8). The attention branch taps f_3 rather than F_b because it retains a higher spatial resolution.

C. Supervised attention branch

a) Attention map.: A lightweight head $g_a(\cdot)$—a 3×3 convolution with BN and ReLU followed by a 1×1 convolution—maps f_3 to a single-channel logit $z = g_a(f_3)$, and a sigmoid $\sigma(\cdot)$ yields the attention map

$$A = \sigma\big(g_a(f_3)\big) \in (0,1)^{1\times\frac{H}{4}\times\frac{W}{4}}. \tag{1}$$

b) Band-mask construction.: For each annotated hyperbola, parameterized by its vertex (x_v, y_v), width w, height h, and thickness t, we rasterize a ground-truth band mask. The central curve is the parabola

$$y_c(x) = y_v + h\left(\frac{x - x_v}{w/2}\right)^2, \qquad x \in \left[x_v - \tfrac{w}{2},\ x_v + \tfrac{w}{2}\right], \tag{2}$$

and the band is the set of pixels within a vertical distance $t/2$ of this curve,

$$B(x,y) = \nVdash \left[\, \left|y - y_c(x)\right| \leq \tfrac{t}{2} \,\right]. \tag{3}$$

The per-image mask is the union over all hyperbolae in the image, down-sampled to the resolution of A and binarized.

c) Attention loss.: The attention map is supervised to match B pixel-wise using a combination of binary cross-entropy and a Dice term. With $A_i, B_i \in [0,1]$ the values at pixel i, N the number of pixels, and a small constant ϵ,

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N}\sum_i \Big[\, B_i \log A_i + (1 - B_i)\log(1 - A_i) \Big], \tag{4}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_i A_i B_i}{\sum_i A_i + \sum_i B_i + \epsilon}, \tag{5}$$

$$\mathcal{L}_{\text{att}} = \mathcal{L}_{\text{bce}} + \mathcal{L}_{\text{dice}}. \tag{6}$$

Crucially, this supervision is applied to a feature map *within the backbone*, not to the detection head, so it steers the encoder to concentrate its response on the hyperbola band region.

D. Attention-guided fusion

To inject the attention prior into the detection pathway, A is average-pooled by a factor of two to align with the bottleneck resolution, $\tilde{A} = \mathrm{AvgPool}_2(A) \in (0,1)^{\,1\times\frac{H}{8}\times\frac{W}{8}}$, concatenated with F_b along the channel axis, and reduced back to C_b channels by a 1×1 convolution $\phi(\cdot)$ (with BN and ReLU):

$$\hat{F} = \phi\big([\, F_b \,;\, \tilde{A}\,]\big) \in \mathbb{R}^{\,C_b\times\frac{H}{8}\times\frac{W}{8}}, \tag{7}$$

where $[\,\cdot\,;\,\cdot\,]$ denotes channel concatenation (yielding $C_b + 1$ channels).

E. Detection head and decoding

Three parallel heads operate on $\hat{F}$ at stride $s = 8$ over an $\frac{H}{s} \times \frac{W}{s}$ grid: an objectness head producing logits O, a size head producing normalized width/height $\hat{S} \in (0,1)^2$ (sigmoid-activated), and an offset head producing sub-cell center offsets $\hat{D} \in \mathbb{R}^2$. For each grid cell (i,j) whose objectness $\sigma(O_{ij})$ exceeds a threshold τ, the box center and size in image coordinates are

$$c_x = \big(j + \mathrm{clip}(\hat{D}^x_{ij}, -0.5, 0.5)\big)\, s, \quad c_y = \big(i + \mathrm{clip}(\hat{D}^y_{ij}, -0.5, 0.5)\big)\, s, \tag{8}$$

$$b_w = \hat{S}^w_{ij}\, W, \qquad b_h = \hat{S}^h_{ij}\, H, \tag{9}$$

giving the box $[\, c_x - \frac{b_w}{2},\ c_y - \frac{b_h}{2},\ c_x + \frac{b_w}{2},\ c_y + \frac{b_h}{2}\,]$. Overlapping candidates are removed by box-level IoU non-maximum suppression.

F. Training objective

Let $P \in \{0,1\}$ be the hard objectness target ($P_{ij} = 1$ at the cell containing an object center, 0 otherwise), $N_+ = \max(\sum_{ij} P_{ij}, 1)$, $N_- = \max(\sum_{ij}(1 - P_{ij}), 1)$, and $\ell_{ij} = \mathrm{BCE}(O_{ij}, P_{ij})$ the per-cell objectness loss. The objectness term balances positives and negatives separately,

$$\mathcal{L}_{\text{obj}} = \frac{1}{N_+}\sum_{ij} P_{ij}\,\ell_{ij} \;+\; \lambda_{\text{noobj}}\,\frac{1}{N_-}\sum_{ij}(1 - P_{ij})\,\ell_{ij}, \tag{10}$$

and the size and offset terms are masked L_1 losses evaluated only at positive cells,

$$\mathcal{L}_{\text{size}} = \frac{1}{N_+}\sum_{ij} P_{ij}\,\big\|\hat{S}_{ij} - S^*_{ij}\big\|_1, \qquad \mathcal{L}_{\text{off}} = \frac{1}{N_+}\sum_{ij} P_{ij}\,\big\|\hat{D}_{ij} - D^*_{ij}\big\|_1, \tag{11}$$

where S^* and D^* are the ground-truth normalized size and sub-cell offset. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{size}} + \mathcal{L}_{\text{off}} + \lambda_{\text{att}}\,\mathcal{L}_{\text{att}}, \tag{12}$$

with $\lambda_{\text{noobj}} = 0.5$ and λ_{att} a weight that balances the attention supervision against the detection terms. Only the attention loss $\mathcal{L}_{\text{att}}$ is added when the supervised branch is enabled; the baseline without attention sets $\lambda_{\text{att}} = 0$ and removes the branch.

V. Experiment

VI. Result and evaluation

A. Dataset

The experiments are conducted on a publicly available GPR dataset for intelligent recognition of subsurface utilities [1]. The dataset consists of labeled GPR B-scan images collected from real urban and industrial environments using ground-coupled antennas operating at 200 MHz and 400 MHz.

B. Result1

C. Result2

D. Result3

E. Result4

F. Result3

G. Result3

H. Result3

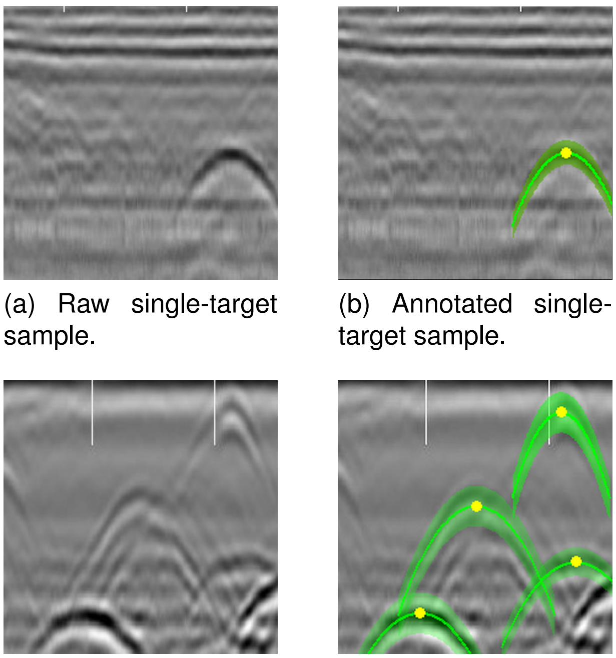
VII. Discussion

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

TABLE I
DETECTION PERFORMANCE (MEAN ± STD OVER 10 SEEDS) AT DIFFERENT TRAINING-DATA RATIOS. "OURS" USES THE ATTENTION CONSTRAINT WITH $\lambda_{\mathrm{att}} = 0.7$. THE BEST F1 PER ROW IS SHOWN IN **BOLD**.

Training ratio	**Baseline**			**Ours** (λ=0.7)		
	P	**R**	**F1**	**P**	**R**	**F1**
0.5	0.811 ± 0.200	0.425 ± 0.094	0.529 ± 0.075	0.855 ± 0.082	0.487 ± 0.070	$\mathbf{0.617 \pm 0.063}$
0.6	0.711 ± 0.213	0.560 ± 0.083	0.600 ± 0.098	0.754 ± 0.254	0.645 ± 0.064	$\mathbf{0.663 \pm 0.131}$
0.7	0.752 ± 0.304	0.679 ± 0.097	0.678 ± 0.207	0.905 ± 0.079	0.782 ± 0.046	$\mathbf{0.836 \pm 0.032}$
0.8	0.843 ± 0.223	0.821 ± 0.071	0.811 ± 0.159	0.909 ± 0.122	0.882 ± 0.041	$\mathbf{0.890 \pm 0.069}$
0.9	0.875 ± 0.238	0.900 ± 0.068	0.867 ± 0.172	0.847 ± 0.223	0.931 ± 0.044	$\mathbf{0.870 \pm 0.165}$



(a) Raw single-target sample.
(b) Annotated single-target sample.
(c) Raw multi-target sample.
(d) Annotated multi-target sample.

Fig. 4. Examples of hyperbola samples and their labeled visualizations.

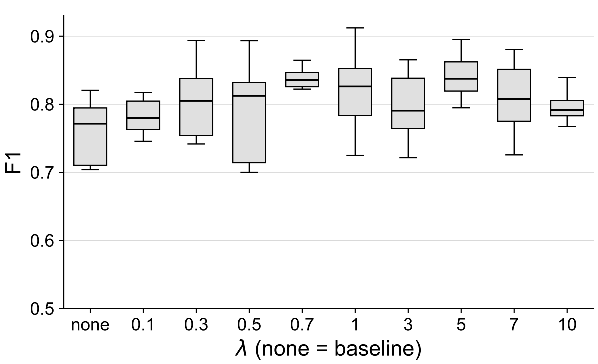


Fig. 5. lambda

TABLE II
COMPARISON OF DETECTION PERFORMANCE WITH DIFFERENT ATTENTION MODULES

Config	Precision	Recall	F1-score
None	0.843 ± 0.223	0.821 ± 0.071	0.811 ± 0.159
SE	0.856 ± 0.195	0.816 ± 0.042	0.821 ± 0.112
CBAM	0.748 ± 0.165	0.819 ± 0.029	0.771 ± 0.097
Non-local	0.844 ± 0.282	0.736 ± 0.225	0.783 ± 0.251
Coord	0.802 ± 0.234	0.836 ± 0.090	0.798 ± 0.172
Ours	$\mathbf{0.909 \pm 0.122}$	$\mathbf{0.882 \pm 0.041}$	$\mathbf{0.890 \pm 0.069}$

APPENDIX
PROOF OF THE ZONKLAR EQUATIONS

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

REFERENCES

[1] A. Mojahid, D. E. Ouai, K. E. Amraoui, K. El-Hami, and H. Ait-benamer, "Intelligent recognition of subsurface utilities and voids: A ground penetrating radar dataset for deep learning applications," *Data in Brief*, vol. 59, p. 111338, 2025.

Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

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